

RESEARCH INTERESTS

Programming Languages, Parallel Computing, Formal Methods, and Software Engineering

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering (Advisor: Prof. Umut Acar)

Aug 2023 — Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering/Mathematics (double major)

Feb 2018 — Feb 2023

- GPA : 4.26/4.3 (ranked 1st in class)
- Entered with *Program for highly Dedicated Students (PHDs)*, presidential scholarship and supporting program offered to one outstanding high school student in the year
- Graduated within 6 semesters (leave of absence for military duty during Jan 2020 — Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 — Feb 2018

- GPA : 4.23/4.3 (ranked 1st in class)
- Funded by Korea Science Academy Fund Scholarship

RESEARCH EXPERIENCES

POSTECH Software Verification Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Kyungmin Bae

Feb 2022 — May 2023

- **Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search**
 - Designed and implemented search heuristic for narrowing-based symbolic reachability analysis of distributed systems
 - Paper accepted to *International Workshop on Formal Techniques for Safety-Critical Systems 2022*

POSTECH Programming Languages Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Sungwoo Park

June 2018 — June 2019

- **MR3**: General purpose distributed execution framework written in Scala
 - Designed and implemented locality-aware task scheduler achieving higher throughput

WORK EXPERIENCES

Kodebox

Seoul, Korea

Software engineer as military service alternative

Jan 2020 — Dec 2021

- **CodeChain Foundry**: Open-source blockchain engine based on composable module system
 - Worked as software engineer
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - CodeChain Foundry is written in Rust
- **Zuzu**: Web-based captable management software
 - Worked as full-stack software engineer
 - Main features implemented include regression testing framework for document generation service, payment, and more
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

PUBLICATION

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search, Byeongjee Kang and Kyungmin Bae, International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022 (doi.acm.org?doi=3563822.3568017)

HONORS AND SCHOLARSHIPS

- Program for Highly Dedicated Students (POSTECH)** *Feb 2018 — Feb 2023*
- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year
- Presidential Science Scholarship** (Ministry of Education of Korea) *Feb 2018 — Feb 2023*
- National science scholarship offered to top freshmen students in the field of science and technology
- Global Leadership Program** (CSE Department, POSTECH) *Sep 2019 — Present*
- Scholarship and research support program for selected students in computer science department
- Domun Scholarship** (POSTECH) *Feb 2019*
- Scholarship for freshmen with highest GPA

TEACHING EXPERIENCES

- Student Mentoring Program** (CSE Department, POSTECH) *Feb 2022 — June 2022*
- Served as tutor to teach Algorithm course
- Intensive Research Program** (POSTECH) *Jan 2019*
- Served as mentor in research experience program for science high school students
- Voluntary Teaching** (POSTECH Volunteering Club) *Jan 2019 — Dec 2019, Sep 2022 — Dec 2022*
- Served as voluntary teacher of math for unprivileged middle school students

EXCHANGE STUDENT & LANGUAGE PROGRAM

- Summer Session Program** *TU Berlin*
Data science course *Aug 2019 — Sep 2019*
- English Program** *University of New South Wales*
English course *Dec 2017 — Jan 2018*
- AwesomeMath Summer Program** *Cornell University*
Mathematics problem solving program for gifted middle and high school students *June 2016 — July 2016*

EXTRACURRICULAR ACTIVITIES & MISCELLANEOUS

- POSTECH Decentralized Autonomous Organization (PDAO)** *POSTECH*
Student association for blockchain development and research *Feb 2022 — Present*
- Served as mentor in blockchain core development team
 - Designed consensus algorithm to be used in PDAO
- Intensive Research Program** *POSTECH*
Research experience program for science high school students - advised by Prof. Jae-Joon Kim *July 2017*
- Learned basic deep learning concepts and researched on techniques to reduce size of DNN
- Research and Education (R&E) Program** *Korea Science Academy of KAIST*
Group research program for juniors - Advised by Dr. Youngchul Jeon *Mar 2016 — Dec 2016*
- Researched properties of *Abelian Ring*, a field of noncommutative ring theory
 - Presented result as school representative in Japan Super Science Fair (JSSF) 2016, at Ritsumeikan High School
- Korea Young Physicists' Tournament (KYPT)** *The Korean Physical Society*
Team-based scientific competition for high school students *June 2017 — Jan 2018*
- Designed and performed experiments, and debated with other teams on the outcome

- Won gold medal

POSCAT

Students club for algorithm study and competitive programming

- Committee member of Postech Programming Contest 2019

POSTECH Student Ambassador

Student-run POSTECH International PR group

- Voluntary service to support exchange students visiting POSTECH

POBBA

Basketball club

- Served as club secretary during the year 2019

POSTECH

Feb 2018 - Dec 2019

International Relations, POSTECH

Feb 2018 — Dec 2018

POSTECH

Feb 2018 — Present

LANGUAGES

Korean (native), English (fluent)

Last updated: Mar 8, 2023